Mo Shirmohammadi

Los Angeles, CA • mshirmoh@usc.edu • github.com/mohosy • mohossy.com

EDUCATION

University of Southern California (USC) *Expected: B.S. Dec 2027*

Computer Science | Presidential Scholar | GPA: 3.90 | Upper Division: 4.0 | Dean's List

- Accepted into USC's Progressive Degree Program (PDP), an optional path to an accelerated M.S. in Computer Science
- Coursework: Computer Networks, Distributed Systems, Operating Systems, Computer Architecture, Algorithms, Probability Theory, Discrete Methods, Software Engineering, Advanced Internetworking (upcoming, Fall 2026)

Pasadena City College (PCC)

A.S., Science, Computer Science | GPA: 3.83 | Courses: Data Structures, Linear Algebra, Multivariable Calc, Differential Equations

EXPERIENCE

Network Engineer, Access Tech Security *January 2025 - June 2026*

- Led network and security deployments at 9+ major client sites (La Salle, Flintridge, Lake Elsinore) as primary liaison between Access Tech and Verkada, owning client relationships, multi-vendor coordination, and end-to-end delivery from site survey to cutover.
- Segmented 45+ client networks (20 to 500 endpoints) with VLANs, inter-VLAN routing, and ACLs across Cisco Catalyst, Ubiquiti UniFi, and FortiGate, isolating cloud-connected IoT security devices from corporate traffic while maintaining a 99.9% uptime SLA.
- Automated configuration backups and inventory tracking via Python (Netmiko) and Excel macros, reducing manual audit time by 15 hours weekly and eliminating configuration drift across 80+ managed switches.
- Implemented SNMP-based monitoring (PRTG) across 35 client sites, establishing proactive alerting that reduced emergency dispatch calls by 60% and cut average incident response time to 45 minutes.
- Managed structured cabling deployments across 23 commercial sites (medical, retail, warehouse), completing 15,000+ ft of Cat6/Cat6a and fiber runs at 100% Fluke DSX-5000 certification pass rate.
- Diagnosed coverage gaps and co-channel interference across 18 high-density sites using Ekahau/AirMagnet surveys, then optimized channel plans and repositioned APs to lift average client signal from -72 dBm to -58 dBm and clear persistent dead zones.
- Standardized rack build procedures and documentation templates, cutting average deployment time from 12 to 6.5 hours per site and post-install support tickets by 40%.

Incoming Software Engineering Intern, Intuitive Surgical *Summer 2026*

- Building cloud-native data infrastructure on AWS and Kubernetes, with distributed pipelines, microservices, and APIs ingesting high-volume telemetry from da Vinci surgical robots and manufacturing systems for downstream analytics and ML.
- Engineering a real-time Kafka/Flink service that flattens high-volume data streams with low latency and low cost, generalizing it into a reusable transformer for other business domains where data flattening is critical.
- Implementing observability and infrastructure as code; hardening services through unit/integration testing, code reviews, and security fixes in an Agile workflow.

Founder & Lead Engineer, LansoAI (Pasadena City College) *April 2023 - March 2025*

- Shipped full-stack AI chatbot serving 30,000+ students with 640+ daily queries; centralized fragmented campus resources into a single conversational interface.
- Engineered Node.js/Express backend for OpenAI GPT-4 integration with custom rate limiting and request caching, cutting response time 50%.
- Built RAG pipeline (vector embeddings, k-NN semantic search) indexing 900+ documents with sub-100ms retrieval; reduced hallucinations 40%.
- Iterated prompts using MongoDB interaction logs as ground truth, boosting answer accuracy from 38% to 66%.
- Established CI/CD pipeline (GitHub Actions, 50+ unit/integration tests) and Redis session caching, cutting deployment errors 60% and API costs 36%.
- Integrated Sentry error logging enabling sub-hour incident response and 99.9% uptime.

Founder, Code Can Bridge *December 2021 - June 2023*

- Founded nonprofit teaching coding to students with learning disabilities; scaled to 3 branches and 80+ students; featured in Forbes Magazine.
- Designed adaptive curriculum and recruited/mentored 10+ volunteer instructors; iterated lesson plans based on student learning outcomes across multiple schools.

PROJECTS

TradeNet Fabric, *Low-Latency Software-Defined Trading Network Simulator* *January 2026 - Present*

- Self-initiated project modeling multi-site trading infrastructure with BGP (iBGP route reflectors, eBGP peering) and OSPF routing across 3 data centers and exchange colocations in EVE-NG.
- Built Python SDN controller ingesting real-time telemetry, calculating optimal paths via latency metrics, and dynamically pushing BGP policy changes.
- Instrumented eBPF-based traffic analysis for per-flow latency measurement with 10μs precision, jitter detection, and packet loss tracking feeding real-time metrics to the SDN control plane for path optimization.
- Developed chaos engineering framework with automated fault injection (link failures, BGP hijacks, route flaps) achieving sub-100ms failover convergence and validating automated recovery.
- Streamlined multi-vendor configuration management (Cisco, Arista, Juniper) with Ansible/Nornir and Jinja2 templates; version-controlled with CI validation.
- Integrated RPKI validation pipeline checking ROA records against BGP announcements; automated alerting and prefix filtering on origin violations.
- Simulated market data distribution via PIM-SM multicast with IGMP snooping; measured jitter variance across redundant paths to optimize feed delivery.

NetShield, *eBPF/XDP Stateful Packet Filter* *May 2026*

- Engineered high-performance Linux firewall using XDP for line-rate packet processing, achieving sub-microsecond filtering before kernel stack entry.
- Programmed stateful connection tracking with BPF maps for TCP/UDP session management, supporting allowlist/denylist policies and rate limiting.
- Created declarative YAML policy engine that compiles rules to eBPF bytecode, enabling version-controlled firewall configurations.
- Exported Prometheus metrics for real-time visibility into drop rates, connection states, and per-rule hit counts.
- Benchmarked against iptables, demonstrating 3x throughput improvement under synthetic load.
- Tested with synthetic DDoS traffic patterns (SYN floods, UDP amplification) validating filtering correctness under adversarial conditions.

IRL Social App, *Founder & Lead Engineer, Real-Time Video Platform (470+ Users)* *March 2024 - January 2026*

- Solo-built and self-funded a video-first social platform ($26K earned working 3 minimum-wage jobs simultaneously), pivoting core features from user interviews to scale school-by-school to 470+ users.
- Architected hybrid real-time layer combining Agora WebRTC (50ms voice/data) with Socket.io for presence and typing indicators.
- Configured NGINX reverse proxy with SSL termination, WebSocket upgrades, and load balancing across multiple Node.js application instances.
- Implemented connection health monitoring with automatic reconnection logic handling network transitions and latency spikes across mobile clients.

SKILLS

Networking & Protocols: ARP, BGP, OSPF, TCP/IP, VLANs, ACLs, SNMP, eBPF, WebRTC

Infrastructure & Tools: Linux (*nix CLI), Cisco IOS/IOS-XE, Arista, Juniper, EVE-NG, Docker, Wireshark, tcpdump, Ansible, Nornir, Netmiko, AWS

Programming: Python, C++, Bash, Git, JSON, GraphQL

Awards: TEDx Speaker (May 2026), Top 10 OpenAI Parameter Competition (2026), 1st Place (Y Combinator Track) at Caltech Hackathon (2026), 1st Place HTCC Honors Conference (2024), 2x Gold PVSA

Publications Featured In: Forbes Magazine, POCIT (People of Color in Tech), PCC Courier, USC SHINE Research